

The Embodiment Shortcut: Action-Following Metrics

Read the Robot, Not the Physics

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Abstract

Video world models for robotics are routinely evaluated by *action following*: an inverse dynamics model (IDM) infers the action from generated video and is compared with the action the generation was conditioned on. The metric is read as physics-sensitive. We show that it is dominated by *embodiment rendering*: when the action is an actuator command, the manipulator’s pose is a near-direct readout of it, so an IDM learns to read the arm and ignore the object. In a controlled MuJoCo push task with a pre-registered analysis, removing the object’s physics entirely (a decoy placed outside every reachable push path) changes the standard metric’s error by only $0.15\times$ its own identifiability floor (95% CI [0.12, 0.19]), and a 16-fold friction sweep leaves it flat at 1.9% of the uninformed baseline. We propose **OG-AF** (Object-Grounded Action-Following), which scores at a delayed horizon after the arm has retracted. OG-AF recovers the action from the settled object (44.7% of baseline versus 99.0% with the decoy), detects physics changes that move the object by 2.5 mm where the standard metric is blind below 80 mm, and separates a world model trained on wrong physics from a correct one with a paired effect of $d = 1.94$ versus 0.46 for the standard metric, whose response is fully accounted for by a no-physics control. All thresholds were frozen and publicly time-stamped before the data existed.

1 Introduction

A video world model conditioned on a robot action should produce video in which that action has its physical consequences. Action following has become a common way to check this [5, 6]: train an IDM on real video, run it on generated video, and measure how well it recovers the conditioning action. A low error is taken as evidence that the model “follows” the action, and implicitly that it gets the interaction right.

We argue that this inference fails for a structural reason. When the action is an actuator command, the manipulator’s configuration after executing it is close to an invertible function of the action. An IDM trained to minimise action-prediction error will exploit that channel, not as a pathology but because it is the optimal solution to the task as posed. The metric then scores how faithfully a world model renders the robot, while being largely blind to whether contact physics was correct. We call this the *embodiment shortcut*.

Contributions. (1) A controlled testbed and estimator that isolate the shortcut: paired INTERACT/DECOY rollouts with identical arm motion and a paired gap G (§2). (2) Pre-registered evidence that the standard metric is insensitive to object physics and that a delayed-horizon variant, OG-AF, is not (§3). (3) A resolution analysis: OG-AF detects millimetre-scale physics changes the standard metric cannot see (§4). (4) A world-model ladder in which OG-AF, but not the standard metric, separates a model trained on wrong physics, including a diagnosed failure of our own first ladder (§5).

2 Setup

Task. A 4-DOF paddle pushes one of three objects (box, sphere, cylinder) in MuJoCo [4], rendered at 224×224 , 20 fps. An action $a = (v, \theta, d)$ (push speed, approach angle, contact offset) is drawn uniformly; we report errors in normalised action units, where the uninformed (uniform-prior) baseline has MAE 0.25. The arm is kinematically clamped to the commanded path, so its motion is identical across conditions by construction; a yaw joint makes $a \mapsto$ arm pose invertible, the situation under study.

Table 1: Experiment E (held-out, pooled over geometries; 611 pairs per seed, 5 seeds). Errors as % of the uninformed baseline; G /floor with 95% CI.

IDM	INTERACT	DECOY	ABSENT	G / floor
A-std (standard)	1.9	2.2	1.5	0.15 [0.12, 0.19]
A-clip-del	2.2	2.5	2.0	0.14 [0.11, 0.16]
A-time	4.1	4.2	3.2	0.02 [-0.02, 0.06]
A-del (OG-AF)	44.7	99.0	98.5	1.21 [1.18, 1.24]

Conditions. For each (a, seed) tuple we simulate INTERACT (object in the push path), DECOY (the same object placed in an annulus proven disjoint from every reachable push corridor, 14 mm clearance, from an independent random stream) and ABSENT (no object). 2000 tuples per geometry give 18 000 rollouts; splits are by tuple so pairs never straddle them.

Horizons. s_{std} is the last frame of the push (arm mid-motion), the horizon the standard metric uses; $s_{\text{del}} = \max(\text{object settled}, \text{arm at rest})$; s_{time} is the same time as s_{del} but with the arm only ϵ -withdrawn (1.5 mm), separating retraction from elapsed time.

IDMs. Architecture A follows the MultiWorld protocol [6]: an ImageNet-initialised ResNet-50 with a non-causal temporal transformer. **A-std** reads a 16-frame clip spanning $[s_0, s_{\text{std}}]$ (VPT-style [1]), the faithful reproduction of the standard metric; **A-del** (OG-AF) reads only the pair (s_0, s_{del}) ; **A-time** reads (s_0, s_{time}) ; **A-clip-del** reads a clip ending at s_{del} . Five seeds each, trained on all three conditions. Architecture B (frozen DINOv3 ViT-L/16 CLS features [3] and an MLP, ten seeds) is reported as descriptive support; DINOv3 was selected because it resolves a 1.7-pixel object shift at $+13.9\sigma$ where DINOv2 [2] manages $+0.55\sigma$.

Estimator. On held-out tuples, $G = \text{mean}_p[\text{err}(\text{DECOY}_p) - \text{err}(\text{INTERACT}_p)]$, the mean of per-pair differences. If a metric uses the object, DECOY should be harder and G large; if it reads only the arm, $G \approx 0$. We express G relative to each IDM’s identifiability floor (its own held-out INTERACT error) and report hierarchical-bootstrap 95% CIs (tuples within seeds, then seeds; 10 000 replicates).

Pre-registration. Thresholds, estimators and decision rules were committed and pushed to a public repository before the confirmatory experiments ran, and a mechanical lock refuses to unseal the analysis otherwise [anonymised repository]. Post-hoc changes are listed in a deviations log; none affects a confirmatory result.

3 The shortcut exists and the standard metric falls for it

State oracles. Before any pixels: a probe on the arm’s state alone recovers the action to 1.0–1.2% of baseline at s_{std} , identically in INTERACT and DECOY, while at s_{del} the arm carries nothing (97–98%) and the settled object still carries signal (11–45%).

Confirmatory results (Table 1). **C1:** the standard metric’s gap is $0.15 \times$ its floor, below the pre-registered threshold $\tau_1 = 0.25$ (CI upper bound 0.19; Holm-adjusted one-sided $p = 0.0002$). Deleting the object’s physics barely matters to it. **C2:** OG-AF recovers the action from the settled object at 44.7% of baseline, below the registered midpoint of floor and baseline (Holm $p = 0.0002$); with the decoy it is at chance (99.0%), as it must be, since at s_{del} the arm is at rest in every condition. **T5:** A-time behaves like A-std ($|\Delta G| = 0.0005$) not A-del (0.135): the effect is driven by retraction, not elapsed time. **T8:** giving the IDM a clip that ends at s_{del} (A-clip-del) re-opens the shortcut, G matching A-std.

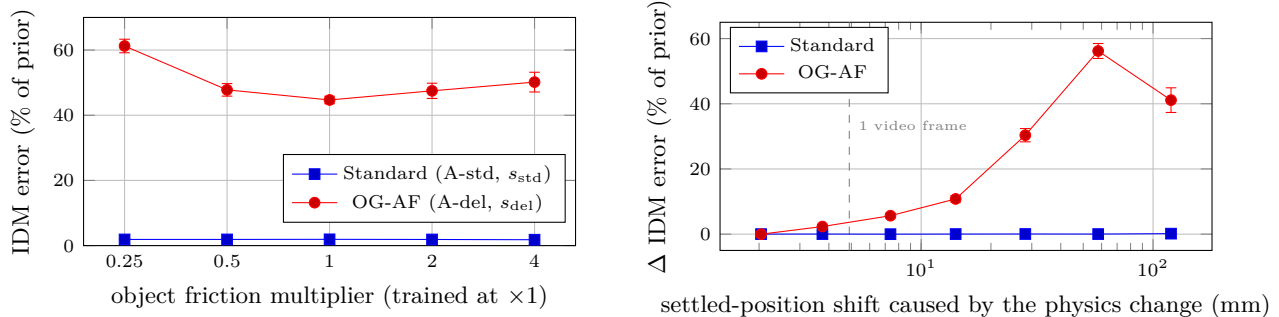


Figure 1: **Left:** the same IDMs, trained at friction $\times 1$, scored on rollouts with changed object friction (held-out tuples, 95% CI). The standard metric stays at 1.8–1.9% while the object’s settled position moves by up to 4 cm; OG-AF rises on both sides of the training value. **Right:** error change versus the physics-induced shift of the object’s settled position, 1 570 (tuple, friction) pairs (post-hoc, exploratory).

4 Resolution: how large a physics error each metric can see

Experiment F scores the trained IDMs on rollouts whose object friction was changed from $\times 0.25$ to $\times 4$ (Fig. 1, left). Because the arm is identical across frictions, the standard metric cannot register the change: it stays at 1.8–1.9% of baseline although the object comes to rest up to 4 cm away. OG-AF’s error rises away from the training friction in both directions (61.2% at $\times 0.25$). (Our pre-registered prediction of a *monotone* response was mis-specified for an IDM trained at $\times 1$; it is reported as failed.)

Pooling every held-out tuple and friction level gives a resolution curve (Fig. 1, right). OG-AF detects physics changes that move the object by ≥ 2.5 mm (half a video frame of motion; $d = 0.64$ at 2.5–5 mm, rising to 3.27 at 40–80 mm). The standard metric shows no detectable change below 80 mm and +0.15% of baseline beyond. OG-AF’s floor is higher (44.7% versus 1.9% of baseline): it trades resolution of the action for sensitivity to the interaction, which is what an action-following metric is meant to measure.

5 World models: catching wrong physics

We train factorised spatio-temporal video transformers (90.5M parameters, AdaLN action conditioning, strictly causal in time, 120k steps) on our corpus and generate the 193 held-out box tuples autoregressively from the first frame. The registered ladder is WM-base at 100%, 30% and 10% of training, a model trained on 10% of the data, and a model trained on wrong physics.

C3(a). On every model, the standard and OG-AF metrics rank the same generated rollouts almost independently (Spearman $\rho \leq 0.28$; threshold < 0.9).

A diagnosed failure. Our first ladder failed its validity gate (ground-truth error monotone across models; $\rho = 0.70 < 0.9$), voiding the between-model claim as registered. The gate measured whole-frame pixel error, which is dominated by how the arm and background are rendered (the object covers $\sim 2\%$ of the frame), and the corruption (friction $\times 3$) was never checked to be large. The validity check fell for the shortcut it was meant to guard against. We registered a second ladder before running it: an object-region ground truth (pixels where the real INTERACT and ABSENT frames differ), and a corruption chosen mechanically as the mildest friction multiplier whose *real* video already exceeds the weakest correct model’s object error ($\times 0.1$).

C3(b). The rebuilt ladder is monotone under object-region error ($\rho = 1.00$; 0.057, 0.059, 0.070, 0.114, 0.162). Paired over the same tuples, the wrong-physics model differs from the correct one by $d = 1.94$ [1.73, 2.22] under OG-AF and $d = 0.46$ [0.37, 0.56] under the standard metric; the registered contrast is $\Delta d = 1.48$ [1.22, 1.79], *supported*. The standard metric’s response is explained by a no-physics control (Fig. 2).

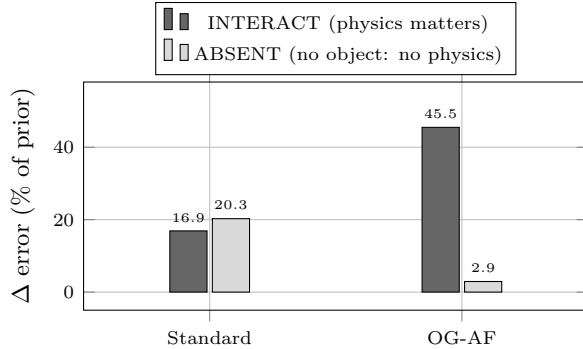


Figure 2: Wrong-physics world model (friction $\times 0.1$) minus the correct WM-base, mean IDM error on the same 193 held-out tuples. On INTERACT the standard metric rises by about as much as on ABSENT, where there is no object and physics cannot matter: it registers the corrupted model’s slightly worse rendering. OG-AF rises sharply on INTERACT and not on ABSENT.

Domain gap. On generated video the standard metric’s error is 56% of baseline for every base model, versus 1.9% on real video, and 61% on the ABSENT control: on generated video it largely measures rendering fidelity. OG-AF sits at its real-video floor (47% versus 44.7%) and at chance on ABSENT.

6 Related work

World-model evaluation increasingly uses action following [6]; a recent diagnosis of whether robotic world models follow actions measures end-effector trajectories, i.e. arm kinematics by design [5]. A position paper argues that IDM-based recovery claims are claims about the whole pipeline [7]; we measure one way the pipeline is entangled. The World Action Verifier [8] assumes a verification subset whose next state depends only on its own state and the action, and proves a sparse inverse model transfers under it; read as an evaluation metric, that condition is object-blindness. Its conditions hold at s_{std} and fail at s_{del} in our data: the guarantee and the confound are the same property used for different purposes.

7 Limitations

Simulation only: one push task, three primitive objects, a kinematically clamped arm, and small world models trained on their own corpus; transfer to real robot data and published world models is the main open question. OG-AF requires a horizon at which the manipulator is at rest, and its floor is higher than the standard metric’s. The world-model evaluation uses 193 held-out box tuples (the registration anticipated 1500). The wrong-physics corpus loses 14–16% of rollouts to the settling validator, so that model also saw less data, which is consistent with its worse rendering on ABSENT. DINO features are blind to sub-degree rotation. The resolution curve is post-hoc. Two pre-registered exploratory predictions failed as worded (friction monotonicity; masking orderings) and are reported as such.

8 Conclusion

Action following, as practised, rewards rendering the robot correctly. Scoring at a horizon where the manipulator no longer encodes the action turns the same pipeline into a measurement of the interaction. We recommend that action-following results report an object-grounded variant and a no-physics control alongside the standard number.

Reproducibility. Code, pre-registrations, deviations log, every result record and the generators for all tables and figures are in [anonymised repository]; one command re-submits the pipeline on a SLURM cluster.

Use of AI assistance. An AI coding agent implemented and audited substantial parts of the pipeline and, at the researchers’ delegation, fixed the open registration decisions (recorded in both registration documents). The authors take responsibility for all decisions and claims.

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